

[Aim: 100/100 in Maths]

PLEQV (L-4)

ATK²:-

Solution (x/y fit values)

Graphical

Algebraic

① substitution

② Elimination

#LP: Check whether the point $(-4, 3)$ lies on both the lines represented by the linear equations $x + y + 1 = 0$ and $x - y = 1$. (CBSE 2024)

$$x + y + 1 = 0$$

$$(-4, 3) \xrightarrow{\text{LHS}} -4 + 3 + 1$$

$$-1 + 1$$

$$\Rightarrow 0 = \text{RHS} \checkmark$$

lie on 1st line

$$x - y = 1$$

$$(-4, 3) \rightarrow \underline{\text{LHS}} -4 - 3$$

$$\Rightarrow -7 \neq \text{RHS}$$

does not lie on 2nd line

2. Elimination method:

जिसको $\frac{y}{x}$ ही eliminate करना है $\rightarrow$ उनके coeff. equal कर दो।

$$1. \underline{x} + 2y = 6 \quad (\times 2)$$

$$\underline{2x} - 5y = 12$$

$$\begin{array}{r} 2x + 4y = 12 \\ 2x - 5y = 12 \\ \hline 9y = 0 \\ y = 0 \end{array}$$

put $\rightarrow$

$$2x - 0 = 12 \quad \checkmark$$
$$x = 6 \quad \checkmark$$

#LP: Solve the following pair of linear equations for x and y using elimination method:

$$\begin{aligned}x + 2y &= 9 \\ y - 2x &= 2\end{aligned}$$

pvt

$$\begin{aligned}x + 2y &= 9 \\ -2x + y &= 2 \quad \times 2 \Rightarrow -4x + 2y = 4\end{aligned}$$
$$\begin{array}{r}x + 2y = 9 \\ -4x + 2y = 4 \\ \hline 5x = 5 \\ x = 1\end{array}$$

$$\begin{aligned}1 + 2y &= 9 \\ 2y &= 9 - 1 \\ 2y &= 8 \\ y &= 4\end{aligned}$$

LP: Solve for x & y: (using elimination)

(i)

$$3x - 4y = 9 \longrightarrow 3x - 4y = 9$$

$$1x - 1y = 3 \times (3) \Rightarrow 3x - 3y = 9$$

$$-y = 0$$

$$y = 0$$

put

$$x - 0 = 3$$

$$x = 3$$

iii

$$3x - 7y = 6 \quad \times 3 \Rightarrow 9x - 21y = 18$$

$$9x + 3y = 1 \quad \longrightarrow \quad 9x + 3y = 1$$

$$-24y = 17$$

$$y = -\frac{17}{24} \quad \checkmark$$

put

$$9x + 3\left(-\frac{17}{24}\right) = 1$$

$$9x - \frac{17}{8} = 1$$

$$9x = 1 + \frac{17}{8}$$

$$9x = \frac{8+17}{8}$$

$$9x = \frac{25}{8}$$

$$x = \frac{25}{72} \quad \checkmark$$

500
(111)

$$2x - 6y = 1 \times (3) \Rightarrow 6x - 18y = 3$$

$$3x + 7y = 2 \times (2) \Rightarrow 6x + 14y = 4$$

$$+32y = +1$$

$$y = \frac{1}{32}$$

put

$$3x + 7\left(\frac{1}{32}\right) = 2$$

$$x = \checkmark$$

(iv)

$$\begin{aligned} 3x - 2y &= 7 \times (3) \Rightarrow 9x - 6y = 21 \\ 7x + 3y &= 1 \times (2) \Rightarrow 14x + 6y = 2 \end{aligned}$$

$$\underline{23x = 23}$$

$$\boxed{x = 1}$$

$$\boxed{(1, -2)} \sim$$

$$7(1) + 3y = 1$$

$$7 + 3y = 1$$

$$3y = 1 - 7$$

$$\cancel{3y} = -6 \Rightarrow$$
$$\boxed{y = -2} \checkmark$$

Special Type $\Rightarrow$ Coefficients interchange

#LP: $217x + 131y = 913$ — (A)

$131x + 217y = 827$ — (B)

(A) + (B)

$$217x + 131y = 913$$

$$+ \quad 131x + 217y = 827$$

$$348x + 348y = 1740$$

$$348(x + y) = 1740 \quad | \quad 5$$

$$\boxed{x + y = 5} \quad \text{--- (I)}$$

(A) — (B)

$$217x + 131y = 913$$

$$- \quad 131x + 217y = 827$$

$$86x - 86y = 86$$

$$86(x - y) = 86 \quad | \quad 1$$

$$\boxed{x - y = 1} \quad \text{--- (II)}$$

put

$$x + y = 5$$

$$x - y = 1$$

⊕

$$2x + 0 = 6 \quad 3$$

$$x = 3$$

$$3 + y = 5$$

$$y = 5 - 3$$

$$y = 2$$

#LP: If $17x - 19y = 53$ and $19x - 17y = 55$, then the value of $(x + y)$ is (CBSE 2022)

- (A) 1
- (B) -1
- (C) 3
- (D) -3

$$\begin{array}{r} \textcircled{+} \\ 17x - 19y = 53 \\ \textcircled{+} 19x - 17y = 55 \\ \hline 36x - 36y = 108 \\ \cancel{36}(x - y) = \cancel{108} \quad 3 \\ \boxed{x - y = 3} \end{array}$$

$$\begin{array}{r} \textcircled{-} \\ 17x - 19y = 53 \\ 19x - 17y = 55 \\ \hline -2x - 2y = -2 \\ \cancel{-2}(x + y) = \cancel{-2} \quad 1 \\ \boxed{x + y = 1} \quad \checkmark \end{array}$$

#LP: Solve:

Number of solutions?

- (A) 0
- (B) 1
- (C) 2 ✓
- (D) Infinite

$$(a^x)^y = a^{xy}$$

$$8^{(x-y)} = (2^3)^{x-y} \Rightarrow 2^{3(x-y)}$$

$$2^{(x+y)} = 64$$

$$8^{(x-y)} = 4$$

$$2^{(x+y)} = 2^6$$

comparing

$$x+y=6$$

$$2^{3(x-y)} = 2^2$$

comparing

$$3(x-y)=2$$

$$3x-3y=2$$

$$64 = 2^6$$

$$\begin{array}{r|l} 2 & 64 \\ \hline 2 & 32 \\ \hline 2 & 16 \\ \hline 2 & 8 \\ \hline 2 & 4 \\ \hline 2 & 2 \\ \hline & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 8 \\ \hline 2 & 4 \\ \hline 2 & 2 \\ \hline & 1 \end{array}$$
$$8 = 2^3$$

$$\begin{array}{l} \rightarrow x + y = 6 \quad \times (3) \Rightarrow 3x + 3y = 18 \\ 3x - 3y = 2 \end{array}$$

$\oplus$

$$3 \cancel{x} = \cancel{20} \quad 10$$

$$x = \frac{10}{3} \quad \checkmark$$

put

$\Downarrow$

$$\frac{10}{3} + y = 6$$

$$y = 6 - \frac{10}{3}$$

$$y = \frac{18 - 10}{3}$$

$$y = \frac{8}{3} \quad \checkmark$$

Module $\Rightarrow$

$1P(I) \rightarrow 1, 2, 3, 4, 5, 6, 9, 11$

$1P(II) \rightarrow 1, 2, 3, 4, 6,$

$2P(I) \rightarrow 2, 3, 4, 6, 8, 9, 12, 13, 20, 22$

$2P(II) \rightarrow 3, 9,$

$2P(III) \rightarrow 4, 5, 6$

THANK YOU COODIESS !!

